

Further experiments

Synthesis of previous experiments up to current results

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Topics

Key points to talk about

Recap

MIDOG++

Results

- Exact-B
- SCEPTRE-HARP

Conclusion

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Recap

Reduced routing risk through composition

What was established on the Camelyon17 dataset:

- Support-NELBO did not translate reliably into Top-1 downstream selection.
- Dense geometric composition exploited complementary CVAE experts.
- Best dense result: **0.814** BACC in the previous experimental setting.

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Last time, I found that compatibility estimation alone was not sufficient. Selecting one expert was brittle, while combining several frozen experts recovered complementary downstream signal. But composition only avoids a potentially wrong choice—it does not solve the routing problem.

Recap

But it did not solve routing

What remained unsolved:

- Composition avoids committing to one potentially wrong expert.
- It does not identify which source expert is compatible with an unseen domain.
- The previous setting offered limited independent domains for rigorous LODO evaluation.
- Move to MIDOG++, with scanner centers as explicit domains.

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I therefore moved to MIDOG++, where scanner centers provide explicit domains for leave-one-domain-out evaluation.

MIDOG++ dataset

Provides more explicit domains for LODO evaluation

Key dataset numbers:

- 492 cases
- 10 observed domains
- 9 eligible LODO domains
- 22.569 image patches

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MIDOG++ dataset

What the dataset contains

- Histopathology data from human and canine tumors
- 7 tumor entities, collected at 4 institutions/origins
- Images acquired using 4 scanner models
- Binary task: mitotic figure vs hard-negative/non-mitotic
- Thesis contract: 11.937 mitotic and 10.632 negative patches

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MIDOG++ dataset

How it is used in this thesis

- Each sample is an annotation-centered 224 x 224 patch
- Patches are represented by frozen 3.840-dimensional Virchow2 embeddings
- Domain definition: tumor type x laboratory/origin x scanner model
- Case-disjoint 45% / 10% / 45% train-validation-test split
- In every LODO fold: 8 source domains → 1 unseen target domain
- One small domain is quarantined, leaving 9 eligible evaluation domains

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MIDOG++ turns routing into a repeated unseen-domain problem: can frozen source experts be selected or composed for a target domain that was unavailable during expert training?

Results

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